their own panel technology, each with its own merit, we'll cover those in the following pages.

Panel technology

There are three primary types of panels available in the market, namely, TN (Twisted Nematic), VA (Vertical Alignment) and IPS (In Plane Switching). TN - Pretty much the oldest and thus the cheapest of the bunch, TN panels form the majority of monitors around. They have faster response times (2 - 5 ms) and high brightness levels. The major drawback is the narrow viewing angle for the panel. Moving out of the view cone results in changes in the perceived colour and viewing from an extreme angle means that you'll barely even make out anything on the screen. The situation is worse when moving vertically as compared to horizontally.

VA - Vertical Alignment panels had even worse viewing angles when they first came out, but within a few years some modifications were made which resulted in greater viewing angles compared to TN panels. VA panels have two major sub-groups - MVA (Multi-domain VA) and PVA (Patterned VA). All these are pretty similar to each other with some focus on one or two aspects. MVA tend to provide better viewing angles and have a higher contrast range as compared to TN and IPS panels whereas PVA panels excel a bit more in the contrast parameter. There are more variants like Premium MVA, Super MVA and Advanced MVA. Every two years or so a new panel technology is invented.

IPS - In Plane Switching panels offer the best colour reproduction and wide viewing angles. This comes with a drawback which is that they are inherently slower, have less brightness and are quite expensive. Wide viewing angles combined with great colour reproduction results in the same picture quality no matter from which angle you look at it. They are thus preferred by people working in multimedia design and animation. Even IPS panels have their minor variants like Enhanced-IPS, Super IPS and Advanced Super-IPS. Each variant improves a bit on the drawbacks of the previous generation.

What's new?

OLED: These are by far the most expensive panels you can get your hands on, the technology is so new that panel sizes are limited which is why they are more common as displays for mobile devices rather than monitors. OLED and it's successor AMOLED boast of the best black levels that can be achieved. This is because in LCD panels the backlight is never switched